

Transformation of plasmids into a bacterial host

Risk assessment

- High-voltage power source
- ▷ Only trained users should operate the electroporator
- ▷ Follow the manufacturer's instructions for operation and shutdown of the electroporation apparatus
- ▷ DO NOT touch the sample chamber or cables during pulsing
- ▷ Wear gloves, safety glasses, lab coat
- Collect and dispose liquid and solid waste after inactivation as REGULATED MEDICAL WASTE



In the event of ELECTRIC SHOCK:
▷ Seek immediate medical attention,
even if symptoms are mild or absent

Reviewed: Feb 19, 2025

This is a bench card. Full protocol available online.

>> Preparation of *E. coli* in logarithmic growth phase

- (1.) Inoculate a single colony into 1–5 mL LB medium. Grow overnight at 37 °C.

Critical: Add antibiotics if needed. For example, to maintain BL21(DE3)pLysS, which carries a plasmid encoding T7 lysozyme, supplement with chloramphenicol. Many laboratory background strains of *E. coli* do not harbor antibiotic resistance markers. In that case, sterilize glassware and plastics used for growth.

- (2.) Inoculate 1.0 vol LB or S. O. B. medium with 5 mL of the overnight culture per liter of final volume. The starting optical density (OD₆₀₀) should be around 0.025 mL⁻¹.

- (3.) Grow the culture at 18 °C, shaking at 150–200 rpm, until it reaches an OD₆₀₀ of 0.4–0.6 mL⁻¹.

Critical: Although growth at 37 °C for 2–3 h is faster, cold growth for at least three doublings is key for high competence. Do not allow the culture to overgrow! Harvest during log phase. Diluting stationary-phase cells is ineffective.

- (4.) Chill the culture on ice for 10 min. Pellet the cells in a refrigerated centrifuge at 1 500–2 000 × *g* for 5–10 min. Carefully decant the supernatant.

A > Chemically competent *E. coli* K-12 and *B* strains for heat shock transformation

- (1.) *Optional:* Resuspend the cells in 0.5 vol ice-cold, sterile 100 mM MgCl₂ solution.
- (2.) Collect by centrifugation as above. Discard the supernatant.
- (3.) Resuspend the cells in 0.5 vol ice-cold, sterile 50 mM CaCl₂ solution. Chill for 15 min on ice.
- (4.) Collect by centrifugation as above. Discard the supernatant.
- (5.) Resuspend in 0.07 vol ice-cold, sterile 50 mM CaCl₂ 15% glycerol solution.

B > Highly efficient chemically competent *E. coli* K-12 and *B* strains for heat shock transformation

□ ◇ R0185 RF1, 200 mL

□ ◇ R0186 RF2, 200 mL

- (1.) Resuspend the cells in 0.3 vol ice-cold RF1. Chill the suspension for 15 min on ice.
- (2.) Collect by centrifugation as above. Discard the supernatant.
- (3.) Resuspend in 0.08 vol ice-cold RF2. Chill for 15 min on ice.

C > **Simple and efficient chemically competent *E. coli* K-12 and B strains for heat shock transformation**

□  R0187 Simple & efficient buffer, 200 mL

- (1.) Resuspend in 0.3 vol ice-cold SEB. Chill the suspension for 10 min on ice.
- (2.) Collect by centrifugation as above. Discard the supernatant.
- (3.) Resuspend in 0.04 vol ice-cold SEB. While swirling the tube, add 0.006 vol DMSO in a drop-wise manner. Chill for 10 min on ice. 

D > **TSS-competent *E. coli* cells**

□  R0188 2 × Transformation & storage medium, 50 mL

- (1.) Resuspend in 0.025 vol cold LB medium. 
- (2.) Add an equal volume of ice-cold 2 × TSS solution. Mix gently by pipetting.

Critical: Do not exceed 30 min incubation on ice. Longer exposure may lower transformation efficiency, possibly due to membrane recovery or PEG toxicity. 

E > **Electrocompetent *E. coli***

- (1.) Resuspend in 0.5 vol ice-cold, sterile, reagent-grade water.
- (2.) Collect by centrifugation as above. Discard the supernatant.
- (3.) Resuspend in 0.2 vol ice-cold, sterile, reagent-grade water.
- (4.) Collect by centrifugation as above. Discard the supernatant.
- (5.) Resuspend in 0.008 vol ice-cold, sterile 10% glycerol. 

>> **Frozen aliquots for long-term storage**

- (1.) Pre-chill all containers that will be used for storage.
Critical: Keep materials on ice and proceed quickly. The shorter this process, the better. 
- (2.) Dispense into 50–200 µL aliquots. 
- (3.) Freeze aliquots in liquid nitrogen or in an ethanol/dry ice bath.
- (4.) Store at –80 °C for up to two years. Freeze-thawing will significantly decrease transformation efficiency.

>> **Preparing the transformation mixture**

- (1.) Thaw cells on ice. Do not warm the tube in your hands, as even brief warming reduces competence. Homogenize the suspension by gently flicking the tube.
- (2.) *Optional:* If necessary, dispense the thawed suspension into additional pre-chilled tubes.
- (3.) To 20–50 µL suspension, add 0.1–100 ng plasmid DNA. The added material should not exceed 0.2 vol of the total volume. Mix the contents by gently flicking the tube.

Transformation of plasmids into a bacterial host

A,B,C > Transformation of chemically competent bacteria

- (1.) Incubate the mixture for 5–30 min on ice.
- (2.) Place the tubes in a water or sand bath of 42 °C for 30–45 s with no shaking. 🔍
- (3.) Place the tubes on ice for 2 min.
- (4.) Add 200–500 µL pre-warmed SOC or LB medium.

D > Transformation of TSS competent bacteria

- (1.) Vortex the suspension. Incubate for 20–30 min on ice. 📖

Critical: Longer incubation time will reduce transformation efficiency. ⬅️

E > Transformation of electrocompetent bacteria

- (1.) Pre-chill the cuvette for electroporation on ice. 📖
- (2.) Pulse with a field strength of 7.5–10 kV/cm, which corresponds to 1.5–2.5 kV for a 0.2 cm cuvette. 🔍 📖

Quality assurance: A successful pulse yields a time constant between 4–8 ms. If arcing occurs, discard both sample and cuvette. 💎

Safety: Turn off the pulse apparatus according to the manufacturer's instructions to fully discharge the capacitors!

- (3.) Add 1 mL pre-warmed SOC or LB medium to the cuvette. Gently pipette up and down to mix. Transfer the suspension to a clean tube for recovery.
- (4.) *Optional:* Rinse the cuvette with 70% ethanol. It can be reused after thorough drying. 📖

>> Selection of transformed clones

- (1.) Incubate the transformed suspension in non-selective growth medium for at least 45–60 min at 37 °C. ⌚ 60 min
- (2.) Spread 0.1–1.0 vol of the recovered suspension onto a selective LB agar plate. Allow the plate to dry and place it upside-down at 37 °C in an incubator. Colonies will be visible after 12–18 h. 📖 🔍 📖

🔗 Recipe (available online) 🔗 Resources (available online) 🔍 Troubleshooting (available online) 📖 Notes (available online)

Lab Protocols — © B. C. Buchmuller. Licensed CC-BY-SA-4.0. Research use; provided as-is, without warranty. Current when printed. Visit <https://benjbuch.github.io/check/?q=ba46f46> or scan the QR code to read the full protocol or to check for updates.

